C Programming Assignment – Question 7

7. Debt Repay

Question

Alice borrows a principal amount at a given annual interest rate for a number of years. Calculate simple interest, total amount, a discount of 2% on the interest, and the final settlement amount. Print each rupee value to two decimal places.

Sample Input

```
100
1
10
```

Sample Output

```
10.00
110.00
0.20
109.80
```

C Program

```
#include <stdio.h>

int main(void) {
    double principal, rate, years;
    double interest, amount, discount, settlement;

    printf("Enter principal amount: ");
    scanf("%lf", &principal);

    printf("Enter annual interest rate: ");
    scanf("%lf", &rate);

    printf("Enter number of years: ");
    scanf("%lf", &years);

    interest = principal * rate * years / 100.0;
    amount = principal + interest;
    discount = interest * 0.02;
    settlement = amount - discount;

    printf("%.2f\n", interest);
    printf("%.2f\n", amount);
    printf("%.2f\n", discount);
    printf("%.2f\n", settlement);

    return 0;
}
```

Explanation

1. Calculate simple interest using principal × rate × years / 100.

2. Add interest to the principal to get the total amount.

3. Calculate the discount as 2% of the interest.

4. Subtract the discount from the total amount to get the final settlement.

5. Print all four values to two decimal places.

Note

For the sample, interest = 10.00, amount = 110.00, discount = 0.20, and settlement = 109.80.